

main.c x

```
1 // C code for linked list merged sort
2 #include <stdio.h>
3 #include <stdlib.h>
4
5 struct Node
6 {
7     int data;
8     struct Node* next;
9 };
10
11 struct Node* SortedMerge(struct Node* a,
12                          struct Node* b);
13 void FrontBackSplit(struct Node* source,
14                   struct Node** frontRef,
15                   struct Node** backRef);
16
17 void MergeSort(struct Node** headRef)
18 {
19     struct Node* head = *headRef;
20     struct Node* a;
21     struct Node* b;
22
23     if ((head == NULL) ||
24         (head->next == NULL))
25     {
26         return;
27     }
28
29     FrontBackSplit(head, &a, &b);
30
31     MergeSort(&a);
32     MergeSort(&b);
33
34     *headRef = SortedMerge(a, b);
35 }
36
```

```

struct Node* SortedMerge(struct Node* a,
                        struct Node* b)
{
    struct Node* result = NULL;
    if (a == NULL)
        return (b);
    else if (b == NULL)
        return (a);
    if (a->data <= b->data)
    {
        result = a;
        result->next =
            SortedMerge(a->next, b);
    }
    else
    {
        result = b;
        result->next = SortedMerge(a, b->next);
    }
    return (result);
}

void FrontBackSplit(struct Node* source,
                  struct Node** frontRef,
                  struct Node** backRef)
{
    struct Node* fast;
    struct Node* slow;
    slow = source;
    fast = source->next;

```

```
int main()
{
    struct Node* res = NULL;
    struct Node* a = NULL;

    push(&a, 15);
    push(&a, 10);
    push(&a, 5);
    push(&a, 20);
    push(&a, 3);
    push(&a, 2);

    // Sort the above created Linked List
    MergeSort(&a);

    printf("Sorted Linked List is: \n");
    printList(a);

    getchar();
    return 0;
}
```



```
Sorted Linked List is:  
2 3 5 10 15 20 |
```